Outlook

[Sigmetrics '26 SRC] Submission #10 "ksentry: A Verified Order-Sensitive..."

From Sigmetrics '26 SRC HotCRP <noreply-sigmetrics26src@hotcrp.com>
Date Wed 4/1/2026 10:52 AM
To Ghimire, Sudrit <aui25@txstate.edu>

Dear author,

The ACM Sigmetrics 2026 Student Research Competition (SRC) (Sigmetrics 2025 SRC) program committee is regret to inform you that your submission #10 has been rejected and will not participate in the SRC.
At same time, we still appreciate the interesting ideas and potential in your work and would like to invite you to present your poster in the student poster session.

* Title: ksentry: A Verified Order-Sensitive Telemetry Accumulator
* Site: https://nam04.safelinks.protection.outlook.com/?url=https%3A%2F%2Fsigmetrics26src.hotcrp.com%2Fpaper%2F2310&data=05%7C02%7Caui25%40txstate.edu%7Ce9b9c43bbb6f4d808fdf08de9006b1a9%7Cb19c134a14c94d4caf65c420f94c8cbb%7C0%7C0%7C639106555611064634%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIwLjAuMDAwMCIsIlAiOiJXaW4zMiIsIkFOIjoiTWFpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=oGDoynSNRt%2FEcaA7KNRMsxLOlQs18VrsNf5%2B4gDg3F8%3D&reserved=0

Visit the submission site for reviews, comments, and related information. Reviews and comments are also included below.

The next steps are:

1) If you decide to present your work at the poster session, please confirm by replying to this email.

2) Register for Sigmetrics. We are in the process of confirming the registration discount for student poster presenters; we will send out details about the registration soon.

3) Prepare a poster to be presented at the poster session during Sigmetrics 2026. We will send out more details closer to the actual conference.

More details can be found here:
https://nam04.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.sigmetrics.org%2Fsigmetrics2026%2Fstudent_activities.html&data=05%7C02%7Caui25%40txstate.edu%7Ce9b9c43bbb6f4d808fdf08de9006b1a9%7Cb19c134a14c94d4caf65c420f94c

8cbb%7C0%7C0%7C639106555611085213%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIwLjAuMDAwMCIsIlAiOiJXaW4zMiIsIkFOIjoiTWFpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=zNk%2BiUjkfCBMr8vPkIlxhms%2FC4lTDp7%2FfuUvk4SXpD8%3D&reserved=0

We look forward to seeing you at Sigmetrics for chatting more about your project.

Contact Emilio Incerto <emilio.incerto@imtlucca.it> or Shaolei Ren <shaolei@ucr.edu> with any questions or concerns.

Emilio Incerto & Shaolei Ren
ACM Sigmetrics 2026 SRC Co-chairs

Review #10B
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Overall merit
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3. Weak accept

Reviewer expertise
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1. No familiarity

Paper summary

Overall the paper presents a nice design, K-Sentry, a compact streaming telemetry accumulator. The author presented a formal bound and a Rust implementation.

Comments for authors

Thank you for submitting to SRC!
The paper has nice theoretical contribution.

Some suggestions:
1) Facts in papers need references to back up. For example, the first sentence in intro, "Modern cloud systems generate high-volume telemetry streams for monitoring and automated control." here you need to cite several papers/news articles to prove that your claim is true.
2) A brief related work paragraph is needed.
3) you have table 1 and table 2 presented but you did not link them in the text
4) Implementation details missing. Where did you implement? Rust? How? Evaluation and experimental set up? You don't need to write a long technical report, but some levels of details are still needed.

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Review #10C

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Overall merit
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3. Weak accept

Reviewer expertise
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1. No familiarity

Paper summary

The problem is practical and the approach is well-scoped. Table 1 makes a clear case for why K-Sentry sits in a useful spot between lightweight checksums. And the prototype results on real Google telemetry traces are convincing. The collision bound analysis is also a welcome addition.

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Review #10D
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Overall merit
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3. Weak accept

Reviewer expertise
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2. Some familiarity

Paper summary

This paper proposes K-Sentry, a constant-memory telemetry accumulator that binds each log value to its position using q^{i^2} mod p, with a Rust implementation mechanically verified in Verus. The paper states refinement properties for the streaming computation, gives a probabilistic collision bound for distinct equal-length traces, and reports prototype experiments on synthetic data and a Google cluster trace showing order-sensitive tamper detection at multi-million-log throughput.

Comments for authors

This is a clear and well-motivated paper, and the formal-verification angle is appealing. I especially liked the emphasis on compositional aggregation and the effort to connect mechanized refinement to a concrete telemetry application.

For improvements, my main suggestion is that the authors could discuss in more details the paper's positioning relative to existing order-sensitive fingerprinting/hash ideas. The paper should explain more

clearly why the quadratic weighting is preferable to more conventional ones. Relatedly, the claims about exact mismatch localization feel somewhat stronger than what is formally established in the current version; it is not fully clear whether localization is a theorem, an experimental observation, or something that requires extra prefix/segment information beyond a single final accumulator value.

Moreover, the evaluation is promising but still fairly light. The paper would be stronger with a broader comparison set, more detail on the cost of modular exponentiation and parameter selection, and a cleaner separation between accumulator overhead and surrounding pipeline overhead.

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Review #10E
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Overall merit
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2. Weak reject

Reviewer expertise
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3. Knowledgeable

Paper summary

This paper presents K-Sentry, an order-sensitive streaming accumulator for telemetry traces, defined as \(A(w;q,p)=\sum_i x_i q^{i^2} \bmod p\). The paper argues that binding each log entry to a position-dependent quadratic weight enables detection of edits, swaps, insertions, deletions, and truncation while retaining constant-memory streaming execution. A key contribution is a mechanized Verus refinement argument for the Rust implementation, covering weight correctness, step correctness, prefix correctness, and compositional aggregation. The paper also states a collision bound of at most \((n-1)^2/p\) for distinct equal-length traces under uniformly random \(q\), and reports prototype throughput of \(3.41\times 10^6\) logs/s on synthetic data and \(4.05\times 10^6\) logs/s on a public Google cluster trace.

Comments for authors

This is a clear, concise, and easy-to-follow SRC submission. The problem is practically relevant, since order-sensitive integrity in telemetry pipelines matters for diagnosis and control, and the paper does a good job motivating why simple checksums or XOR summaries are insufficient. The strongest part of the work is the formal-methods angle: connecting a mathematical accumulator specification to executable Rust through Verus is a meaningful systems contribution, especially in a short student-research format. I also found the compositional aggregation property well motivated for distributed telemetry settings. The paper is readable, focused, and better structured than many submissions of this length.

My main concern is that the paper currently overstates its integrity and tamper-detection story relative to the formal guarantee it actually establishes. As written, the core result is a randomized collision bound

for a fingerprint-like construction under uniformly chosen \(q\). That is not the same thing as a cryptographic integrity guarantee against an adaptive adversary. The paper repeatedly uses language such as 鈥淪ttacks detected,鈥 鈥淭amper detection,鈥 and 鈥淔orensics/compliance,鈥 but it does not specify whether \(q\) is secret, how it is chosen, whether it is reused across streams, or what the attacker is assumed to know. Without an explicit adversarial model, the construction reads much more like an order-sensitive polynomial fingerprint than an authenticator. I think this distinction matters a lot, and the claims should be sharpened accordingly.

My second major concern is that the algorithmic design choice is under-motivated. The paper uses quadratic exponents \(q^{i^2}\), but does not explain why this is preferable to the standard linear polynomial fingerprint \(\sum_i x_i q^i\), which is already order-sensitive and, at least at first glance, seems to admit a tighter collision bound of order \((n-1)/p\) rather than \((n-1)^2/p\). If the quadratic indexing is essential, the paper needs to explain what property it enables that a linear construction does not. Right now, this choice feels somewhat arbitrary, which weakens the novelty argument.

A related issue is the paper鈥檚 repeated claim of 鈥渆xact mismatch localization.鈥 The formal refinement statements are about weight, step, prefix, and compositional correctness, while the experiments report localization behavior such as a single-bit tamper at index 9 and a swap localized to boundary indices \(\{0,18\}\). However, the paper never explains what localization algorithm is actually used, what extra information it requires, what its complexity is, or whether localization is part of the verified story or merely an empirical behavior of the prototype. From the final accumulator state alone, one can test equality, but not obviously recover the location of a mismatch. If localization requires checkpoints or intermediate states, that should be stated explicitly and reconciled with the constant memory narrative. As currently written, the localization claim is too strong for the amount of technical detail provided.

I also think the verification boundary should be described more carefully. The paper does acknowledge that the root-counting lemma used for the collision bound is outside the mechanized proof, which is good, but this weakens the verified headline somewhat. For an SRC paper this is not fatal, yet it does mean the submission should be more precise about what is fully verified and what remains trusted mathematical reasoning.

The evaluation is also fairly thin for the breadth of the claims. The correctness examples are tiny (\(n=19\)), the real-trace experiment uses 100,000 entries, and the performance comparison is mostly against a rolling-hash baseline that is about 55x faster. The paper says this is a tradeoff for stronger guarantees, but it does not really show when that tradeoff is operationally attractive. I would have liked to see a more systematic scaling study, a breakdown of where the overhead comes from, and a clearer discussion of how acceptable the reported collision probability is when many streams or windows are checked in production. The reported 10.8 GB memory usage is also attributed to pipeline overhead without enough detail to interpret it.

Finally, the related-work positioning is too sparse. The paper should engage more directly with polynomial fingerprinting and rolling-hash literature, authenticated streaming mechanisms, and verified implementations of integrity-related primitives or authenticated data structures. My sense is that the mechanization angle may be the most novel part of the work, but the current draft does not frame that contribution as clearly as it could.

Overall, I think this is a promising student project with a nice verified-systems flavor, but the current version needs a much clearer threat model, a more convincing justification for the quadratic

construction, a precise explanation of localization, and a deeper evaluation. My recommendation is weak reject in its current form, though I do think the project could become a solid SRC presentation with a more careful and less overstated writeup.

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